DATA ITEM DESCRIPTION

Form Approved
OMB No. 0704-0188

1. TITLE	2. IDENTIFICATION NUMBER
NUCLEAR SURVIVABILITY PROGRAM PLAN	DI-NUOR-80156A

3. DESCRIPTION / PURPOSE

3.1 The Nuclear Survivability Program Plan shall describe how the contractor's Nuclear Survivability Program shall be conducted. It describes operations, procedures, design, analyses, tests, and management activities to be performed to meet the specified nuclear survivability requirements.

(continued on page 2)

4. APPROVAL DATE	5. OFFICE OF PRIMARY RESPONSIBILITY (OPR)	6a. DTIC REQUIRED	6b. GIDEP APPLICABLE
900109	A/LABCOM/SLCHD-NW		

7. APPLICATION / INTERRELATIONSHIP

7.1 Application. When a solicitation or contract contains a requirement for nuclear survivability, this Data Item Description (DID) shall be listed on the Contract Data Requirements List (DD Form 1423)

7.2 This DID contains the format and content preparation instructions for the data product gen-

(continued on page 2)

8. APPROVAL LIMITATION	9a. APPLICABLE FORMS	9b. AMSC NUMBER
		A4867

10. PREPARATION INSTRUCTIONS

10.1 Reference documents: The applicable issue of the documents cited herein, including their approval dates and dates of any applicable amendments, notices and revisions, shall be as specified in the contract.

10.2 General: The Nuclear Survivability Program Plan shall specify the methods and techniques for incorporating nuclear survivability into design, development, and equipment integration, and for conducting a comprehensive nuclear survivability validation program. It shall include a discussion of the procedures and methods to implement hardness maintenance and surveillance of the deployed equipment. The program plan shall be prepared consistent with the information contained in HDL-TR-1882-1, HDL-SR-85-3, and HDL-CR-81-015-1, and shall include the information as described below.

10.3 Specifics described in the plan:

a. The plan shall describe in detail the specific approaches to be utilized to achieve nuclear survivability. This discussion shall include electrical and mechanical hardening design features and any operational or procedural methods planned.

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11. DISTRIBUTION STATEMENT

Distribution Statement A: Approved for public release; distribution is unlimited.

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3. DESCRIPTION/PURPOSE (continued)

3.2 This plan provides the basis for the Government's determination that the contractor's Nuclear Survivability Program will meet contractual requirements in a cost-effective manner and is applicable to development and non-development programs.

7. APPLICATION/INTERRELATIONSHIP (continued)

erated by the specific and discrete task requirements as delineated in the contract

7.3 Nuclear Weapons Effects on Army Tactical Systems, Vol. I, Overview, HDL-TR-1882-I, provides the philosophy and guidance for design, analysis, and evaluation of Army systems' nuclear survivability in supporting the Program Plan preparation.

7.4 AR 70-60, Nuclear Survivability of Army Materiel, contains the Army policy applicable to nuclear survivability. HDL-CR-81-015-1 and HDL-SR-85-3 are reference documents that provide background information to support the preparation of this Program Plan.

7.5 MIL STD 1388-1A and 2A contain the Army policy applicable to Logistic Support Analysis for materiel acquisitions and contain guidance on the annotation and recording of hardness-unique requirements.

7.6 All HDL documents are available from the Defense Technical Information Center.

7.7 This DID supersedes DI-NUOR-80156.

10. PREPARATION INSTRUCTIONS (continued)

b. The plan shall detail the combination of analyses and tests required to demonstrate survivability. A discussion of design margins to be employed in the analysis shall be included.

c. The program plan shall describe the nuclear survivability test program to be performed under the contract. Testing at each level of integration (piece part, circuit, etc.) shall be discussed in terms of need, simulators, test levels, dosimetry, instrumentation, etc.

d. The plan shall include a discussion of the management of the survivability program. The duties and responsibility of primary point of contact shall be identified and discussed. The relationship between the survivability manager and the program manager shall be detailed.

e. The program plan shall include a functional and operational description of the end item being developed or acquired under the respective contract. This description shall address the key subsystem elements in sufficient detail to enable the government program manager to understand the nuclear survivability program approach.

f. The plan shall present a schedule of critical survivability program milestones related to the overall program schedule. This time-phased schedule shall indicate initiation, review, and completion for each key nuclear survivability task.

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g. The plan shall present a general discussion of the integrated logistic support management approach to field a survivable system, including design, analysis, test, and documentation.

h. The plan shall provide the estimated labor hours, the caliber of labor and the costs for material, supplies, travel, computer, and simulation facilities.

10.4 Designs, analyses, tests, and evaluations described in the plan:

a. The plan shall delineate analytical efforts that supplement or replace testing, to include identification of computer codes, and the intended applications of these analytical techniques.

b. The plan shall delineate piece part, component, and subsystem tests to be performed for each specified nuclear environment, with details, such as sample size, test methods, test instrumentation, parameters to be characterized and relationships to analytical efforts.

c. The plan shall delineate system level tests to be performed, including methods of extrapolation from test environments to the threat environments, the rationale for simulator choices and relationships to analytical efforts.

d. The plan shall describe anticipated test simulators and test equipment to be employed, test configurations, test article orientation, exposure levels and test data to be obtained. (Reference HDL-SR-85-3)

e. The plan shall describe the survivability philosophy to include the basis for selection of piece parts, materials, device technologies, circuit and mechanical designs; the specific materials, technologies and designs rejected because of nuclear survivability considerations; assumptions concerning the system design margins; planned operational fixes such as cycling power and any assumptions concerning the system operation, function, deployment or configuration that has been used in developing the Nuclear Survivability Plan. There shall be particular emphasis on design philosophy for hardness critical items (HCI) and mission critical equipments. The plan shall describe the configuration control approach to control HCIs and HCPs.

f. The plan shall describe the approach to the incorporation of nuclear survivability issues and considerations into the appropriate integrated logistic support documentation as early as possible in the program. The integrated logistic support implications shall include operator procedures, maintenance, repair and overhaul procedures, personnel skill levels, spare parts, repair tools, provisioning and training needs. The procedures of MIL-STDs 1388-1A and 2A shall be followed regarding the proper annotation of HCIs and HCPs for the Logistic Support Analysis and the Logistic Support Analysis Record.

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g. The plan shall delineate the projected requirements, driven by the nuclear survivability of the system, for custom or hardened parts, materials, or components; the basis of need for these parts; and the impacts of these requirements on the program and system design, cost, operation, function and deployment.

h. If the prime contractor or any of the subcontractors are supplying nondevelopment or proprietary materials, components, or equipments under this contract, this plan must detail how the nuclear survivability of these materials, components or equipments shall be evaluated, assured, maintained and documented during all acquisition phases.

i. The plan shall identify the nuclear survivability trade-offs that shall be performed against other system requirements such as schedule, cost, mobility, reliability, producibility and integrated Logistic support.

10.5 Risk analysis and identification described in the plan:

a. Nuclear survivability areas of high risk and uncertainty, including the impact on cost, performance, weight, and MANPRINT shall be identified.

b. Assumptions, conclusions, and reasons used in risk analysis and identification shall be delineated.

c. Actions to be taken by the contractor to minimize the impact of risk and uncertainties identified shall be delineated.

d. When failure modes and effects analysis (FMEA) or failure modes and effects criticality analysis (FMECA) is required elsewhere in the contract, the plan shall indicate how the impact of the survivability degradation risks will be integrated.

10.6 Management interface controls described in the plan:

a. Interfaces between the Nuclear Survivability Program and other related programs, such as engineering design, reliability, maintainability, logistic support, configuration management, production engineering, and producibility programs shall be described.

b. Methodology and procedures by which the prime contractor shall ensure that any appropriate nuclear survivability program is executed in subcontracts shall be described.

c. Actions to be taken to ensure nuclear survivability (to include validation, assurance and maintenance) for materiel to be procured elsewhere, including any non-development or proprietary items shall be delineated.

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d. Special requirements to be included in source or product selection shall be delineated.

e. The management strategy for mission essential items which do not meet system or materiel survivability requirements shall be described. This should also be reflected in risk identification (10.5)

10.7 Program management described in the plan:

a. Responsibilities and authorities of the Nuclear Survivability Program Manager and the nuclear weapons effects experts assigned to the program shall be described.

b. Management procedures and controls to assure the successful application and accomplishment of nuclear survivability requirements shall be described.

10.8 Format: The plan shall be in the contractor's format.

10.9 Additional information:

The plan shall provide any additional information necessary to adequately delineate the Nuclear Survivability Program.

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